

Notes: Scroll down to the end to see codes details.

Five almost identical jupyter notebooks were created for this project and only one is shown here.

Implementation of Paper: Mixed-integer optimization approach to learning association rules for unplanned ICU transfer

@CicilyWu

1. Introduction

1.1 Problem Definition

Patients with heavy disease and unexpected deterioration need to be sent to ICU after their visits in Emergency Care. However, it is hard to predict if a person should be sent to ICU and which decreases the work efficiency for clinicians' work. From the beginning, emergency physicians treat patients and make decisions based on CT, symptoms and other showing information. But sometimes patients were sent to ICU due to emergencies which were not quite predictable. It is necessary and urgent to accurately forecast if a person needs to be sent there. The population who were sent to ICU unexpectedly shows a higher mortality rate than expected transfer. Therefore, it is important to figure out rules between patients' other conditions and ICU transfer decisions.

Under this circumstance, a decision making tool should be designed to help clinicians figure out patients real conditions and potential probabilities of sending to ICU. If machine learning methods should be used to deal with this problem, the technology of supervised learning is selected. Generally speaking, two groups of classification results were clear: the person should be sent to ICU or not.

1.2 Data Description

Based on study, nearly eighty percent of inappropriate ICU transfer can be preventable. In recent studies, 204936 ED patients' data was collected. It ruled out more than 56000 people who got trauma problems, around 930 people who got obstetric problems, injury/suicide people and people under 18. And from this, we only studied patients who are admitted to ICU. Overall number of studied patients is 1049 including 313 transferred patients and 736 non-transferred patients.

The dataset was collected by a Taiwan teaching hospital between 1/1/2007 to 12/31/2010. Around five thousands The dataset is composed of binary data of whether or not if a patient has certain health problems. Patients were divided into four groups based on their main reason they visited the Emergency Department. Four groups are: infections, cardiovascular/respiratory disease, gastrointestinal disease and neurological diseases.

The dataset includes final result (case) and other clinical related variables.

2. Proposed Methods

2.1 Association Rules

The research used association rules method as the first step to find out hidden rules between ICU condition and other patients' information.

Associations rules was first invented because of market analytics that they want to find out the correlations between customer goods. Minimum support and minimum confidence are two important indicators to evaluate the accuracy of rules selected. The higher these two number, the better the results show. Lift is another important index related to support and confidence. Lift value =1 means non-correlation between items selected. On the contrary, lift larger than 1 shows a strong correlation between conditions items and result items. Large and complex dataset must exist many strong different rules, in which case it is critical to set values for support and confidence to gain better rules results.

In this project, we set different minimum support and minimum confidence value for 4 subgroups and one all-data group based on paper's default setting.

As an example, we need to gain 3 matrices, setting them as matrix a, b and c. Matrix a is the dataset we got directly from hospital. The columns are final decision (case=1 or 0) and other related information. Each record represents one patient's healthcare data. Matrix b represents the rules we generated through association rules method. In this case, we need to use binary data to show features selection for different rules. Matrix c is a combination mapping result for matrix a and b. By mapping first two matrices, it directly shows us which patients were covered by these selected rules and which are not.

For creating the matrix c, we firstly transpose matrix b. Then we use matrix a to minus matrix b. If any component of the result is less than zero. It means such patient is not covered.

2.2 Optimization Model

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$$(ARSOM) \min \alpha \sum_{j=1}^m y_j + \beta \sum_{k=1}^K z_k + \gamma \sum_{i \in I^+} x_i - \lambda \sum_{i \in I^+} x_i, \quad (2)$$

$$\text{s. t. } \sum_{k \in K} c_{ik}^+ z_k \geq x_i \quad \forall i \in I^+, \quad (3)$$

$$\sum_{k \in K} c_{ik}^- z_k \leq M_i x_i \quad \forall i \in I^-, \quad (4)$$

According to the strong association rules we found before, there are too many rules redundant, which means that a large number of feature combinations are redundant. If all of them are used, the calculation time is exponential, and the cost of the hospital to use those rules to make predictions is also high. We want to choose as few rules as possible, as few features as possible, and at the same time be able to cover as many unplanned transfer patients as possible (case =1). For patients with case = 0, because they are ready for transfer, we don't have to spend effort to predict them.

So we set gamma, beta, alpha = 1. If the patients with unplanned transfer are not covered, the situation can be quite dangerous, and thus a medical accident may occur, so we set lamda equals 999. constraint(3) means that if a patient is covered, at least one rule will cover him. Constraint (4) and (5) mean that if a planned transfer patient is covered by a rule, there will be some punishment. Since if we make effort on such patients, it is a waste. Constraint (6), (7), (8) have been completed according to the association rules, we can remove them in this phase. In addition, big M equals to the number of rules generated in phase one.

We use gurobi to get solutions of the Mixed Integer Linear problem for different subgroups, the optimal rules are obtained to minimize our cost. Based on these rules and the features of rules, we are able to predict new patients. Then we used the confidence of selected rules to get the probability, which will be used in gaining AUC latter.

2.3 Cross Validation

Cross-validation is a technique in which we train our model using the subset of the data and then evaluate using the complementary subset of the data. In cross validation, the data we use is usually split into training data and test data. The training set contains a known output and the model learns on this data in order to be generalized to other data later on. We have the test dataset (or subset) in order to test our model's prediction on this subset. Here we split our data into n subsets, and train on n-1 one of those subset. We hold the last subset for test and repeat for each of the subsets. Here we use k fold cross validation in which we split our data into k different subsets (or folds). We use k-1 subsets to train our data and leave the last subset (or the last fold) as test data. We then average the model against each of the folds and then finalize our model. After that we test it against the test set.

AUC - ROC curve is a performance measurement for classification problem at various thresholds settings. ROC is a probability curve and AUC represents degree or measure of separability. It tells how much model is capable of distinguishing between classes. Higher the AUC, better the model is at predicting 0's as 0's and 1's as 1's. By analogy, Higher the AUC, better the model is at distinguishing between patients with disease and no disease.

For the project, since there are two phases. We need to extract five data sets of 5-fold cross validation. Then apply previous steps in each part. For each subgroup, we define some functions to help us run fast, since we have to go through every step five times in each notebook.

3. Experimental Results

3.1 Data Analysis

1. Data Visualization

Two types of charts shown here. One is correlation matrix showing correlations between different variables. Due to the binary data type, the relationships are not so obviously but some patterns can still be found. Heatmap makes it easier to read and understand. Bar charts also been used to show the distribution of each class in each variable. These two visualizations were applied to 10 datasets in total including subgroup1-4, subgroup_all and corresponding cross-validation versions. Only Subgroup_1 and Subgroup_All are shown in this report as examples.

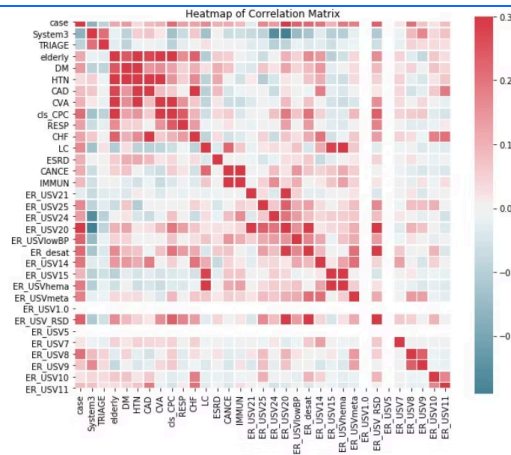


Figure 3 Bar Chart for Subgroup_1 (Part)

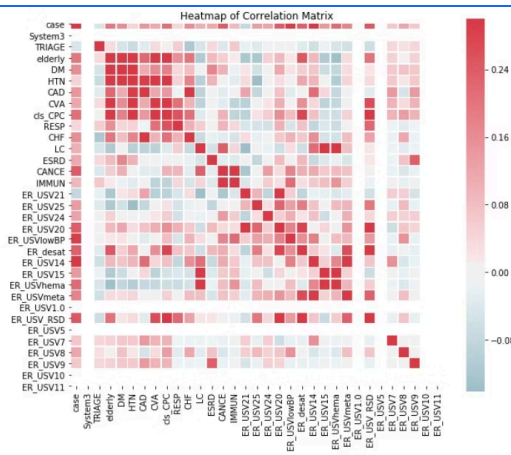
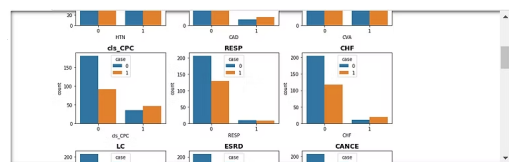


Figure 4 Bar Chart for Subgroup_All

2. Data Processing

Original data provided are binary type and relatively clean. Only instead of containing both “0” and “1”. The reason to do this is that calculation process slow.

3.2 Rules Results



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[‘Hematological dysfunction’],
 [‘Altered mental status’],
 [‘Diabetes’, ‘Respiratory compromise’],
 [‘Hypertension’, ‘Respiratory distress’],
 [‘Coronary artery disease’, ‘Congestive heart failure history’],
 [‘Cerebral vascular disease’, ‘Respiratory failure history’],
 [‘Cerebral vascular disease’, ‘End-stage renal disease’],
 [‘Abnormal white blood cell counts’, ‘Respiratory compromise’],
 [‘Diabetes’, ‘Coronary artery disease’, ‘SIRS’],
 [‘Diabetes’, ‘Heart rate > 130’, ‘SIRS’]

Totally: 13 Rules

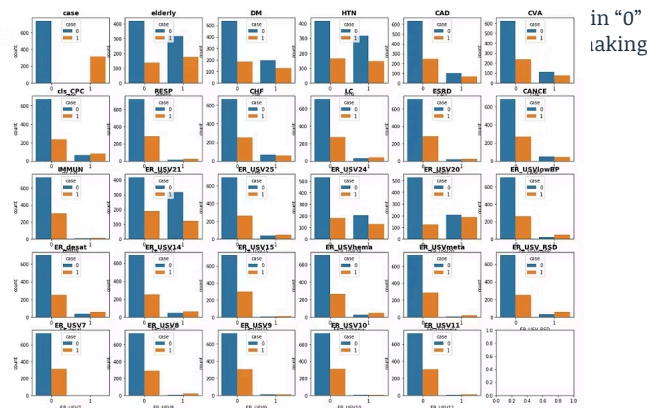
Selected Rules for Subgroup_2: 15 rules

[‘Heart rate > 130’],
 [‘Hypotension’],
 [‘Respiratory distress’],
 [‘Diabetes’, ‘Hematological dysfunction’],
 [‘Liver cirrhosis history’, ‘SIRS’],
 [‘Liver cirrhosis history’, ‘Hematological dysfunction’],
 [‘> 65years’, ‘Cancer’, ‘Abnormal white blood cell counts’],
 [‘> 65years’, ‘Abnormal white blood cell counts’, ‘SIRS’],
 [‘Diabetes’, ‘Hypertension’, ‘Cerebral vascular disease’],
 [‘Diabetes’, ‘Liver cirrhosis history’, ‘Cancer’],
 [‘Hypertension’, ‘Abnormal white blood cell counts’, ‘SIRS’],
 [‘Liver cirrhosis history’, ‘BT >38 C’, ‘Abnormal white blood cell counts’],
 [‘Abnormal white blood cell counts’,
 ‘Liver dysfunction’,
 ‘Hematological dysfunction’]

Totally: 15 Rules

Selected Rules for Subgroup_3: 11 Rules

[‘Cerebral performance category’],
 [‘Respiratory failure history’],
 [‘End-stage renal disease’],
 [‘Hematological dysfunction’],
 [‘Metabolic dysfunction’],
 [‘Chest pain’].



in “0”
 taking

['BT >38 C', 'Abnormal white blood cell counts'],
 ['SIRS', 'Respiratory compromise'],
 ['> 65years', 'Coronary artery disease', 'Cerebral vascular disease']]
 Totally: 11 Rules

Selected Rules for Subgroup_4: 6 rules

[['Respiratory compromise'],
 ['Altered mental status'],
 ['Cerebral vascular disease', 'Cerebral performance category'],
 ['Cerebral performance category', 'Congestive heart failure history'],
 ['Heart rate > 130', 'Abnormal white blood cell counts'],
 ['Cerebral performance category', 'BT >38 C', 'SIRS']]
 Totally: 6 Rules

Selected Rules for Subgroup_All

[['Metabolic dysfunction'],
 ['Altered mental status'],
 ['Diabetes', 'Respiratory distress'],
 ['Cerebral vascular disease', 'Respiratory distress'],
 ['Cerebral performance category', 'Cancer'],
 ['Cerebral performance category', 'Renal dysfunction'],
 ['Liver cirrhosis history', 'Abnormal white blood cell counts'],
 ['Liver cirrhosis history', 'SIRS'],
 ['End-stage renal disease', 'SIRS'],
 ['BT >38 C', 'Hypotension'],
 ['Abnormal white blood cell counts', 'Hypotension'],
 ['Abnormal white blood cell counts', 'Respiratory compromise'],
 ['Abnormal white blood cell counts', 'Respiratory distress'],
 ['SIRS', 'Hypotension'],
 ['SIRS', 'Renal dysfunction'],
 ['SIRS', 'Hematological dysfunction'],
 ['Renal dysfunction', 'Hematological dysfunction'],
 ['Liver dysfunction', 'Hematological dysfunction'],
 ['> 65years', 'Diabetes', 'Heart rate > 130'],
 ['> 65years', 'Diabetes', 'Respiratory compromise'],
 ['> 65years', 'Hypertension', 'Respiratory failure history'],
 ['Diabetes', 'Congestive heart failure history', 'SIRS'],
 ['Diabetes', 'Cancer', 'SIRS'],
 ['Hypertension'],
 ['Cerebral performance category'],
 ['Congestive heart failure history'],
 ['Hypertension', 'Respiratory compromise', 'Renal dysfunction']]
 Totally: 27 Rules

3.3 Cross Validation

For ARSOM model, the dataset was divided into five parts evenly to run 5-fold cross validation. Also, they were also run under other three different machine learning models which are Logistic Regression (LR), LASSo and Decision Trees(DT). They are implemented with python and installed packages have been used. The AUC result for each subgroup under different models are shown as below. The performance of ARSOM model is better than other three models.

4. Conclusion

In this project, rules were selected in each subgroup classified by several diseases by using association rules under python. A small-size rules set was generated by optimization model afterwards. Finally, cross validation has been used to make the result more accurate and AUC scores were generated under different models. We can reach the conclusion that this new model called ARSOM has the best performance among all models, which indicated this could be a good method to identify potential ICU patients.

Chou, C. A. , Cao, Q. , Weng, S. J. ,
icu transfer.

Groups\Models	ARSOM	LR	LASSO	DT
SUBGROUP1	0.83±0.03	0.8±0.02	0.79±0.01	0.69±0.03
SUBGROUP2	0.84±0.03	0.77±0.03	0.78±0.01	0.62±0.03
SUBGROUP3	0.79±0.02	0.54±0.03	0.56±0.01	0.53±0.04
SUBGROUP4	0.76±0.02	0.79±0.16	0.73±0.04	0.68±0.02
SUBGROUP_all	0.78±0.04	0.79±0.10	0.79±0.02	0.61±0.03

Table 1 AUC Result in each subgroup under four models

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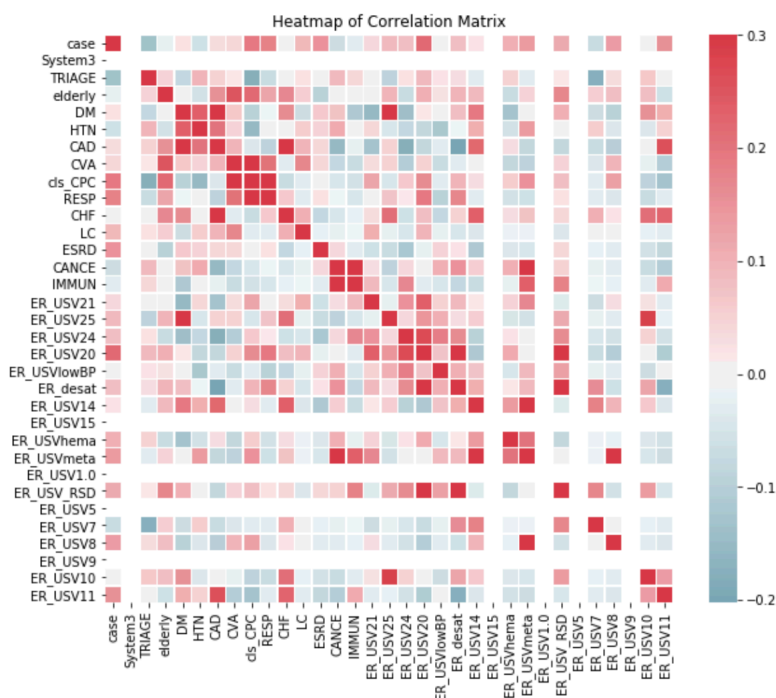
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Healthcare ICU Unplanned Transfer

```
In [6]: # Heatmap
corr = df1.corr()
plt.figure(figsize = (10,10))
cmap = sns.diverging_palette(220, 10, as_cmap=True)
sns.heatmap(corr, xticklabels=corr.columns.values, yticklabels=corr.columns.values, cmap=cmap, vmax=.3,
            center=0, square=True, linewidths=.5, cbar_kws={"shrink": .82})
plt.title('Heatmap of Correlation Matrix')
```

Out[6]: Text(0.5, 1, 'Heatmap of Correlation Matrix')



```
In [7]: # make df2 as df1's copy in order to keep the original version datasets
df2=df1.copy()
#drop System3 and TRIAGE because TA said we won't use these two columns
#Also delete four columns below sine they only contain "0"
df2.drop('System3', axis=1,inplace = True)
df2.drop('TRIAGE', axis=1,inplace = True)
df2.drop('ER_USV11', axis=1,inplace = True)
```